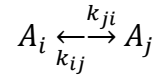


*Wei (Princeton) and Kuo(1969) developed a scheme to identify lumps of species in a large reaction network.

*Consider n species $A_1, A_2, \dots, A_n$ participating in the following reaction under isothermal conditions



*Mole balance for species A_j can be written as

$$\frac{dC_j}{dt} = k_{j1}C_1 + k_{j2}C_2 + \dots + k_{j,j-1}C_{j-1} - \sum_{i=1, j}^n k_{ij}C_j + k_{j,j+1}C_j + \dots + k_{jn}C_n \quad (108)$$

$$\underline{C} = \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_n \end{bmatrix}, \underline{K} = \begin{bmatrix} -\sum_{i=1}^n k_{i1} & k_{12} & \dots & k_{1n} \\ k_{21} & -\sum_{i=1,2}^n k_{i2} & \dots & k_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ k_{n1} & k_{n2} & \dots & -\sum_{i=1}^{n-1} k_{in} \end{bmatrix}$$

So that

$$\frac{d\underline{C}}{dt} = \underline{K}\underline{C} \quad (109)$$

*Lumping of reaction species implies species concentration be defined by $\widehat{C}_1, \widehat{C}_2, \dots, \widehat{C}_p$ lumps where each element is a lump of species

*For example:

$$C_1 + C_2 + \dots + C_{r_1} = \widehat{C}_1 \quad (Lump \ 1)$$

$$C_{r_1+1} + C_{r_1+2} + \dots + C_{r_1+r_2} = \widehat{C}_2 \quad (Lump \ 2)$$

...

$$C_{r_1+r_2+r_{p-1}+1} + \dots + C_{r_1+r_2+r_p} = \widehat{C}_p \quad (Lump \ p)$$

$$r_1 + r_2 + \dots + r_p = n$$

This can be represented using a lumping matrix

$$\underline{M} = \begin{bmatrix} 1 & 1 & \dots & 1 & 0 & 0 & \dots & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 1 & 1 & \dots & 1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 0 & 0 & \dots & 0 & \dots & 1 & 1 & \dots & 1 \end{bmatrix}$$

Such that

$$\underline{MC} = \underline{\hat{C}} \quad (110)$$

Multiply eqn (109) by $\underline{M}$ we get

$$\underline{M} \frac{dC}{dt} = \underline{MKC} \rightarrow \frac{d\hat{C}}{dt} = \underline{MKC} \quad (111)$$

We seek a reduced kinetic model of the type

$$\frac{d\hat{C}}{dt} = \underline{\hat{K}}\hat{C}$$

Substitute (110)

$$\frac{d\hat{C}}{dt} = \underline{\hat{K}}\underline{MC} \quad (112)$$

Compare (111) and (112)

$$\underline{MK} = \underline{\hat{K}}\underline{M} \quad (113)$$

This is the condition of lumpability.

*We start with hypothesizing various lumps based on chemical nature or number of carbons.

*We can lump group species that are similar in reactivity (or which interconvert via quasi-equilibrated steps)

*After proposing various lumps, we need to check whether the lumps are proper or improper.

* $\underline{\hat{C}}$ is a proper representation of C if $\frac{d\hat{C}}{dt}$ is independent of the concentration of individual components within a lump.

*All the terms in $\underline{M}$ are 1 or 0.

*Each column has only one term.

Ex: $\underline{\hat{C}} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $\underline{C}$ is a proper lump. $\underline{\hat{C}}_1 = C_1 + C_2$, $\underline{\hat{C}}_2 = C_3$

While $\underline{M} = \begin{bmatrix} 1 & 0.5 & 0 \\ 0 & 0.5 & 1 \end{bmatrix}$ would lead to improper lumping.

*Accuracy and ineffectiveness of lumping schemes

*How far we can lump before the lumpy model becomes inaccurate.

*Ex: A 10-lump model for catalytic cracking. Read “lumping analysis” pdf on blackboard.

*5. Determining rate constants in first order systems.

*Rate constants are the essential parameters for the reactor models.

*Rate constant can be determined from experiments.

Ex: consider a first order reaction $A \rightarrow B$

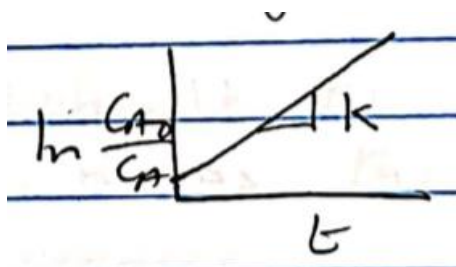
Mole balance of A can be written as

$$\frac{dC_A}{dt} = -kC_A, \quad C_A(0) = C_{A,0}$$

$$\text{Integrate, } C_A = C_{A,0} \exp^{-kt}$$

$$\ln \frac{C_A}{C_{A,0}} = -kt, \quad \ln \frac{C_{A,0}}{C_A} = kt \quad (114)$$

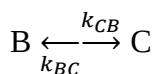
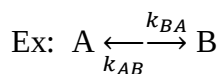
Plot of $\ln \frac{C_{A,0}}{C_A}$ vs t would yield rate constant k



In case of n^{th} order kinetics

$$\left(\frac{1}{C_A^{n+1}} - \frac{1}{C_{A,0}^{n+1}} \right) = \frac{kt}{n} \quad (115)$$

*For a single reaction, it is straight forward to estimate rate constants. However, for a multiple reactions, the rate constants are difficult to estimate.



$$\frac{dC_A}{dt} = k_{AB}C_B - k_{BA}C_A$$

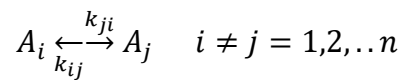
$$\frac{dC_B}{dt} = k_{BA}C_A + k_{BC}C_C - k_{AB}C_B - k_{CB}C_B$$

$$\frac{dC_C}{dt} = k_{CB}C_B - k_{BC}C_C$$

*Difficult to solve! But can be solved numerically!

*It gets increasingly difficult to estimate rate constants at the number of reactions increases.

*Our objective is to identify a method to measure rate constants of all reactions.



k_{ji} is a rate constant of reaction producing A_j from A_i

The rate of reaction: $r_{ji} = j_{ji} C_i$ (116)

There are $2 \times nC_2 = n(n-1)/2$ reactions possible between n species.

The mole balance for an isothermal constant volume, well-stirred batch reaction

$$\frac{dC_j}{dt} = \sum_{i=1}^m \alpha_{ij} r_i = \sum_{i,j=1}^n r_{ji} - \sum_{i,j=1}^n r_{i,j} = \sum_{i=1}^n k_{j,i} C_i - \sum_{i=1,j}^n k_{i,j} C_j$$

$$\frac{d\underline{C}}{dt} = \underline{K} \underline{C} \quad (109)$$

Sum all the elements of $\underline{C}$

$$\frac{d}{dt} \sum_{j=1}^n C_j = 0$$

$$\sum_{j=1}^n C_j = C \quad (116)$$

Total concentration is constant.

Let x_j be the mole fraction of A_j $x_j = \frac{C_j}{C}$, such that $\sum_{j=1}^n x_j = 1$ (117)

Divide eq (109) by C

$$\frac{d\underline{x}}{dt} = \underline{K} \underline{x} \quad (118)$$

$$\underline{X} = [x_1 \ x_2 \ x_3 \ \dots \ x_n] \text{ and } K = [k_{ij}] \text{ with } k_{ii} = -\sum_{j=1, i}^n k_{j,i} \quad (119)$$

At equilibrium

$$\frac{dx}{dt} = 0 \rightarrow \underline{K} \underline{x}^* = \underline{0} \quad (120)$$

From (119)

$$k_{ii} + \sum_{j=1, i}^n k_{j,i} = 0 \rightarrow \sum_{j=1}^n k_{j,i} = 0$$

$\underline{K}$ is a singular matrix $\rightarrow |\underline{K}|=0$

Solve $\underline{K} \underline{x}^* = \underline{0}$ to obtain $\underline{x}^*$. Note that $\lambda = 0$ is one of the eigen values of $\underline{K}$

*From the principle of microscopic reversibility or detailed balance.

$$A_i \xrightleftharpoons[k_{ij}]{k_{ji}} A_j$$

$$k_{ij}x_j^* = k_{ji}x_i^* \quad (121)$$

Define a diagonal matrix D

$$\underline{D}^{1/2} = \begin{bmatrix} \sqrt{x_1^*} & 0 & 0 & \dots & 0 \\ 0 & \sqrt{x_2^*} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & \sqrt{x_n^*} \end{bmatrix}$$

$$\underline{D}^{-1/2} = \begin{bmatrix} 1/\sqrt{x_1^*} & 0 & 0 & \dots & 0 \\ 0 & 1/\sqrt{x_2^*} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1/\sqrt{x_n^*} \end{bmatrix}$$

Such that

$$\underline{D}^{1/2} \underline{K} \underline{D}^{-1/2} = \begin{bmatrix} k_{11} & k_{12}\sqrt{x_2^*} & 0 & \dots & 0 \\ k_{21}\sqrt{x_1^*} & k_{22} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & k_{nn} \end{bmatrix}$$

From (121)

$$k_{ij} \sqrt{\frac{x_j^*}{x_i^*}} = k_{ji} \sqrt{\frac{x_i^*}{x_j^*}} \rightarrow \hat{k}_{ij} = \hat{k}_{ji}$$

$$\underline{D}^{-1/2} \underline{K} \underline{D}^{1/2} = \begin{bmatrix} \hat{k}_{11} & \hat{k}_{12} & \hat{k}_{13} & \cdots & \hat{k}_{1n} \\ \hat{k}_{21} & \hat{k}_{22} & \hat{k}_{23} & \cdots & \hat{k}_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \hat{k}_{n1} & \hat{k}_{n2} & \hat{k}_{n3} & \cdots & \hat{k}_{nn} \end{bmatrix} = \underline{\hat{K}}$$

$\underline{\hat{K}}$ is a summetric matrix

Theorem 1: If $\underline{D}^{-1/2} \underline{K} \underline{D}^{1/2} = \underline{\hat{K}}$ for any non zero $\underline{D}$, then $\underline{K}$ vs $\underline{\hat{K}}$ are similar matrices with identical eigenvalues.

Let $\lambda_0 \underline{z}_0, \lambda_1 \underline{z}_1, \lambda_2 \underline{z}_2, \dots, \lambda_n \underline{z}_n$ be the eigenvalues and eigenvectors of $\underline{K}$. And $\lambda_0 \underline{\hat{z}}_0, \lambda_1 \underline{\hat{z}}_1, \lambda_2 \underline{\hat{z}}_2, \dots, \lambda_n \underline{\hat{z}}_n$ be the eigenvalues and eigenvectors of $\underline{\hat{K}}$.

Theorem 2: The eigenvector of a symmetric matrix are orthogonal to each other. $\underline{z}_i \perp \underline{z}_j$
 $\underline{z}_j$ are the eigenvector of $\underline{\hat{K}}$

$$\underline{\hat{K}} \underline{\hat{z}}_j = \lambda_j \underline{\hat{z}}_j$$

$$\underline{D}^{-\frac{1}{2}} \underline{K} \underline{D}^{\frac{1}{2}} \underline{\hat{z}}_j = \lambda_j \underline{\hat{z}}_j \text{ (from 122)}$$

$$\underline{K} \underline{D}^{1/2} \underline{\hat{z}}_j = \lambda_j \underline{D}^{1/2} \underline{\hat{z}}_j$$

But

$$\underline{K} \underline{z}_j = \lambda_j \underline{z}_j$$

$$\underline{\hat{z}}_j = \underline{D}^{1/2} \underline{z}_j \text{ (123)}$$

Applying theorem 2

$$0 = \underline{\hat{z}}_j^T \underline{\hat{z}}_k \quad (j \neq k) = \underline{\hat{z}}_j^T \underline{D}^{-\frac{1}{2}} \underline{D}^{-\frac{1}{2}} \underline{\hat{z}}_k$$

$$\underline{\hat{z}}_j^T \underline{D}^{-1} \underline{\hat{z}}_k = 0 \text{ (124)}$$

Theorem 3: Any vector can be expressed a linear combination of independent vectors.

$$\underline{x} = \alpha_0 \underline{z}_0 + \alpha_1 \underline{z}_1 + \alpha_2 \underline{z}_2 + \cdots + \alpha_{n-1} \underline{z}_{n-1} = \sum_{j=0}^{n-1} \alpha_j \underline{z}_j \text{ (125)}$$

To calculate coefficient α_j multiply both sides by $\underline{\hat{z}}_j^T \underline{D}^{-1}$

$$\begin{aligned}\hat{\underline{z}}_j^T \underline{D}^{-1} \underline{x} &= \alpha_0 \hat{\underline{z}}_j^T \underline{D}^{-1} \underline{z}_0 + \alpha_1 \hat{\underline{z}}_j^T \underline{D}^{-1} \underline{z}_1 + \alpha_2 \hat{\underline{z}}_j^T \underline{D}^{-1} \underline{z}_2 + \cdots + \alpha_{n-1} \hat{\underline{z}}_j^T \underline{D}^{-1} \underline{z}_{n-1} \\ &= \alpha_j \hat{\underline{z}}_j^T \underline{D}^{-1} \underline{z}_j\end{aligned}$$

$$\alpha_j = \frac{\hat{\underline{z}}_j^T \underline{D}^{-1} \underline{x}}{\hat{\underline{z}}_j^T \underline{D}^{-1} \underline{z}_j} \quad (126)$$

Note

$$\alpha_0 = \frac{\hat{\underline{z}}_0^T \underline{D}^{-1} \underline{x}}{\hat{\underline{z}}_0^T \underline{D}^{-1} \underline{z}_0}$$

$$\alpha_0 = \frac{\underline{x}^* \underline{D}^{-1} \underline{x}}{\underline{x}^* \underline{D}^{-1} \underline{x}^*}$$

Substitute (126) in (125)

$$\underline{x} = \underline{x}^* + \sum_{j=0}^{n-1} \left\{ \frac{\hat{\underline{z}}_j^T \underline{D}^{-1} \underline{x}}{\hat{\underline{z}}_j^T \underline{D}^{-1} \underline{z}_j} \right\} \quad (127)$$

We want to solve (118)

$$\frac{d\underline{x}}{dt} = \underline{k} \underline{x} \quad , \quad \underline{x}(0) = \underline{x}^i$$

Premultiply both sides by $\underline{z}_j^T \underline{D}^{-1}$

$$\frac{d}{dt} (\underline{x}^T \underline{D}^{-1} \underline{z}_j) = \underline{x}^T \underline{K}^T \underline{D}^{-\frac{1}{2}} \underline{D}^{-\frac{1}{2}} \underline{z}_j = \underline{x}^T \underline{K}^T \underline{D}^{-\frac{1}{2}} \hat{\underline{z}}_j = \underline{x}^T \underline{D}^{-\frac{1}{2}} \hat{\underline{K}} \hat{\underline{z}}_j$$

$$\underline{D}^{-\frac{1}{2}} \hat{\underline{K}} \underline{D}^{-\frac{1}{2}} = \hat{\underline{K}} \rightarrow \underline{D}^{-\frac{1}{2}} \underline{K}^T \underline{D}^{-\frac{1}{2}} = \hat{\underline{K}}$$

Take transpose again

$$\frac{d}{dt} (\underline{z}_j^T \underline{D}^{-1} \underline{x}) = \lambda_j \underline{z}_j^T \underline{D}^{-1} \underline{x}$$

$$\underline{z}_j^T \underline{D}^{-1} \underline{x} = \underline{z}_j^T \underline{D}^{-1} \underline{x}^i e^{-\lambda_j t} \quad (128)$$

Substitute (128) in (127)

$$\underline{x} = \underline{x}^* + \sum_{j=0}^{n-1} \left\{ \frac{\hat{\underline{z}}_j^T \underline{D}^{-1} \underline{x}^i e^{-\lambda_j t}}{\hat{\underline{z}}_j^T \underline{D}^{-1} \underline{z}_j} \right\} \quad (129)$$

Theorem 4: symmetric matrix with negative diagonal elements have all eigenvalues negatives.

Arrange eigenvalues in decreasing order

$$\lambda_0 > \lambda_1 > \lambda_2 \dots > \lambda_{n-1}$$

Such that

$$e^{\lambda_0 t} > e^{\lambda_1 t} > e^{\lambda_2 t} > \dots > e^{\lambda_{n-1} t}$$

Choose a large enough t, so that $e^{\lambda_2 t}, \dots, e^{\lambda_{n-1} t} \sim 0$

$$\underline{x} = \underline{x}^* + \left(\frac{\underline{z}_i^T \underline{D}^{-1} \underline{x}^i}{\underline{z}_j^T \underline{D}^{-1} \underline{z}_j} \right) \underline{z}_i e^{\lambda_1 t} = \underline{x}^* + \alpha_i \underline{z}_i e^{\lambda_1 t}$$

$$\rightarrow (\underline{x} - \underline{x}^*) = \alpha_i \underline{z}_i e^{\lambda_1 t}$$

$$\frac{\underline{x} - \underline{x}^*}{\underline{z}_i} = \alpha_i e^{\lambda_1 t}$$

$$\ln(\underline{x} - \underline{x}^*) = \ln(\alpha_i \underline{z}_i) + \lambda_1 t$$

Conduct experiment with initial composition $\underline{x}^i$ and measure $\underline{x}$ vs t

Plot $\underline{x}$ vs t and draw a tangent at $\underline{x}^$. The slope of the tangent is λ_1 and intercept is $\alpha_i \underline{z}_i$

*Now with $\underline{z}_i$ vs λ_1 known define new initial composition

$$\underline{x}^{i,2} = \underline{x}^{i,1} - \frac{\underline{z}_i^T \underline{D}^{-1} \underline{x}^{i,1}}{\underline{z}_i^T \underline{D}^{-1} \underline{z}_i} \underline{z}_i \quad (130)$$

Substitute (130) in (129)

$$\underline{x} = \underline{x}^* + \left(\frac{\underline{z}_i^T \underline{D}^{-1}}{\underline{z}_i^T \underline{D}^{-1} \underline{z}_i} \right) \left(\underline{x}^{i,1} - \frac{\underline{z}_i^T \underline{D}^{-1} \underline{x}^{i,1}}{\underline{z}_i^T \underline{D}^{-1} \underline{z}_i} \underline{z}_i \right) + \sum_{j=2}^{n-1} \left\{ \frac{\underline{z}_j^T \underline{D}^{-1} \underline{x}^{i,2}}{\underline{z}_j^T \underline{D}^{-1} \underline{z}_j} \right\}$$